

Membership-inference audit report

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1 run(s) collected

run directory	type	algorithm	dataset*	roles	curves
three_way_cifar10_neggrad_plus	level3	neggrad_plus	cifar10	3	18

* dataset is read from the run directory name, not the summary (the summaries do not record it).

Level-3 U-RMIA (strong attack, train-then-unlearn references)

1 run(s)

Comparison across runs

Two-sided AUC on the unlearned model, one column per scorer.

run	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online	unl test	unl forget
three_way_cifar10_neggra	0.5422	0.5410	0.6378	0.6391	0.6154	0.6113	0.6820	0.8100

three_way_cifar10_neggrad_plus

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar10

Configuration

algorithm : neggrad_plus
forget_size : 500
params : epochs=2, learning_rate=0.002, forget_coef=0.5, retain_coef=1.0, forget_loss_cap=1.16

Accuracy

role	test_acc	forget_acc
original	0.7040	n/a
unlearned	0.6820	0.8100
retrained	0.7160	0.6980

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.6766	0.6776	0.7079	0.7508	0.7672	0.7635
unlearned	0.5422	0.5410	0.6378	0.6391	0.6154	0.6113
retrained	0.4809	0.4808	0.5077	0.4926	0.4930	0.4957

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.6766	0.6776	0.7079	0.7508	0.7672	0.7635
unlearned	0.5422	0.5410	0.6378	0.6391	0.6154	0.6113
retrained	0.5191	0.5192	0.5077	0.5074	0.5070	0.5043

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0020	0.0020	0.0560	0.0340	0.0380	0.0160
unlearned	0.0000	0.0000	0.0220	0.0060	0.0280	0.0080
retrained	0.0000	0.0000	0.0040	0.0000	0.0000	0.0020

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0100	0.0060	0.0680	0.0980	0.0860	0.0500
unlearned	0.0040	0.0040	0.0540	0.0380	0.0480	0.0340
retrained	0.0060	0.0100	0.0220	0.0080	0.0080	0.0100

Retain-set leakage

members = 1000-point retain sample, 500 unseen non-members. delta = score(unlearned) - score(original) per retain point; frac_inc = fraction whose delta is positive; ctrl_d = the same delta for retrained vs original.

scorer	frac_inc	d_mean	d_med	ctrl_d	auc_orig	auc_unl	tpr1_orig	tpr1_unl
urmia_offline	0.6950	-0.0135	0.0036	-0.0068	0.7899	0.7747	0.2490	0.2310
urmia_online	0.7170	-0.0154	0.0051	-0.0076	0.6600	0.6423	0.0000	0.0000
ulira	0.4520	-0.3279	-0.0746	1.2204	n/a	n/a	n/a	n/a
conf_global	0.4150	-0.0301	-0.0002	0.0024	0.6775	0.6606	0.0110	0.0070

Three-way test (trained / unlearned / never)

16 references per point, fit from references only. Cells are the fraction of each group assigned to each world; member = trained or unlearned; advertised = below the never baseline by 2.0 sigma. collateral (never world shift from unlearning other points): mean -0.0632, forget -0.0384, unseen -0.0880.

target/group	trained	unlearned	never	member	advertised
original/forget	0.6000	0.2660	0.1340	0.8660	0.0000
original/unseen	0.2680	0.2720	0.4600	0.5400	0.0400
unlearned/forget	0.4360	0.3040	0.2600	0.7400	0.0900
unlearned/unseen	0.2880	0.2660	0.4460	0.5540	0.0500
retrained/forget	0.3400	0.2260	0.4340	0.5660	0.0380
retrained/unseen	0.3080	0.2260	0.4660	0.5340	0.0300

binary balanced acc	forget vs unseen
original	0.6630
unlearned	0.5930
retrained	0.5160

Positive control

dead_scorers : none
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7672, urmia_online=0.7635, ulira=0.7079, conf_global=0.6766, mentr_global=0.6776, ulira_mentr=0.7508

Timing

Loading dataset took 9.87769 seconds
Model loading/training took 4221.3 seconds
Preparing signals took 6.22278 seconds
Auditing took 48.1 seconds
Retain-leakage analysis took 2.0 seconds
Three-way analysis took 1.4 seconds
Total runtime: 4289.25967 seconds

three_way_cifar10_neggrad_plus — log-scale ROC

